

A Topological Ansatz for the Leptonic CP Phase: G₂-Manifold Betti Numbers and the DUNE Experiment

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Abstract

We propose a phenomenological ansatz that correlates the leptonic CP-violating phase (δ_{CP}) with the topological invariants of G₂-holonomy manifolds arising in M-theory compactifications. By identifying a specific vacuum candidate defined by the Betti number pair ($b_2 = 13, b_3 = 7$), we obtain a parameter-free prediction of $\delta_{CP} \approx 197.8$, consistent with the 1σ interval of current global fits (NuFit 6.0). We explicitly acknowledge that this framework constitutes a heuristic protocol based on pattern recognition—what we term “Keplerian phenomenology”—rather than a rigorous derivation from the M-theory effective action. Its scientific utility lies entirely in its empirical risk: providing a non-adjustable target for the forthcoming DUNE experiment. If the prediction fails, the ansatz is cleanly falsified; if it succeeds, the “coincidence” will require theoretical explanation.

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1 Introduction: The Need for Rigidity

The Standard Model of particle physics, despite its extraordinary predictive success, contains approximately 19 free parameters whose values are determined experimentally rather than derived from first principles. Among these, the parameters of the neutrino sector—particularly the leptonic CP-violating phase δ_{CP} —remain among the least constrained.

Current global fits from the NuFit collaboration [1] report:

$$\delta_{CP}^{\text{exp}} = 194 \pm 25 \quad (1\sigma, \text{Normal Ordering}) \quad (1)$$

The forthcoming Deep Underground Neutrino Experiment (DUNE) is expected to measure δ_{CP} with a precision of approximately ± 5 by 2030, providing a decisive test for any framework that makes specific predictions for this parameter.

In this paper, we explore the possibility that continuous physical constants may emerge from the discrete topology of extra dimensions. Specifically, we focus on compactifications of M-theory on manifolds with G_2 holonomy, which provide a natural setting for realizing $\mathcal{N} = 1$ supersymmetry in four dimensions.

1.1 Methodological Position: Keplerian Physics

We position this work as an exercise in **Keplerian phenomenology**: identifying empirical patterns in data before the underlying dynamical laws are fully understood. Just as Kepler’s elliptical orbits preceded Newton’s derivation of gravitational dynamics, we propose a numerical pattern that—if confirmed—would demand theoretical explanation.

We emphasize that:

- The formulas presented are **ansätze**, not derivations
- The framework makes **falsifiable predictions**, not postdictions
- Confirmation would motivate, not prove, the underlying hypothesis

2 Geometric Framework

2.1 Established Context (Category A)

The geometry of G_2 manifolds is characterized by their Betti numbers b_k . In the context of M-theory compactifications, the following relations are well-established in the literature [2, 3]:

Established Results

- **Chiral Supermultiplets** (b_3): The number of massless chiral fermions (matter fields) in the 4D effective theory is determined by the third Betti number. Additionally, b_3 determines the dimension of the moduli space of torsion-free G_2 structures.
- **Vector Supermultiplets** (b_2): The number of abelian $U(1)$ gauge fields is determined by the second Betti number, associated with co-associative 4-cycles.

These are mathematical facts, not conjectures.

2.2 The Proposed Topology

We investigate the specific configuration:

$$b_2 = 13 \quad (\text{co-associative 4-cycles}), \quad b_3 = 7 \quad (\text{associative 3-cycles}) \quad (2)$$

Important Caveat

The pair $(b_2 = 13, b_3 = 7)$ lies below the typical range of explicitly constructed compact G_2 -holonomy manifolds. Joyce’s original examples [2] have $b_3 \geq 10$, and Twisted Connected Sum (TCS) constructions [4] typically produce even larger values.

We propose this topology as a **phenomenological target**; its geometric realization remains an open mathematical problem.

3 The Topological Ansatz

3.1 The Conjectures (Category C)

We propose that the CP-violating phase is governed by the ratio of the vacuum’s topological complexity to its effective instanton level.

Conjecture 1 (Instanton Level):

$$k_{CS} = b_2 \cdot b_3 = 13 \times 7 = 91 \quad (3)$$

Conjecture 2 (Phase Formula):

$$\delta_{CP} = 2\pi \cdot \frac{b_3^2 + 1}{k_{CS}} = 2\pi \cdot \frac{50}{91} \quad (4)$$

Result:

$$\delta_{CP} = 197.802 \approx 198 \quad (5)$$

This value lies within the 1σ confidence interval of NuFit 6.0 (169–219).

3.2 Theoretical Audit: Acknowledged Gaps

To maintain scientific integrity, we explicitly acknowledge that the following elements **lack first-principles derivation**:

Unresolved Theoretical Questions

1. **The identification** $k_{CS} = b_2 \cdot b_3$: While Chern-Simons theory on G_2 manifolds exists [5], no published work identifies the CS level as the product of Betti numbers.
2. **The numerator** $N = b_3^2 + 1 = 50$: The specific structure of this expression is heuristic. We do not claim it counts any known topological object.
3. **The selection of** $(13, 7)$: Among all possible Betti number pairs, why this one corresponds to our universe (if it does) is unexplained.

These gaps define a roadmap for future theoretical work, contingent on experimental confirmation.

4 Phenomenological Stability

A key question is whether the prediction $\delta_{CP} \approx 198$ is robust or fine-tuned. We examine neighboring topologies:

Table 1: Sensitivity of δ_{CP} to Betti number variations

(b_2, b_3)	k_{CS}	$b_3^2 + 1$	δ_{CP} (°)	NuFit 1σ Compatible?
blue!10 (13, 7)	91	50	197.8	Yes
(12, 7)	84	50	214.3	Yes
(14, 7)	98	50	183.7	Yes
(13, 6)	78	37	170.8	Marginal
(13, 8)	104	65	225.0	Yes
(11, 7)	77	50	233.8	Yes
(15, 7)	105	50	171.4	Marginal
(13, 5)	65	26	144.0	No
(13, 9)	117	82	252.3	No
(10, 7)	70	50	257.1	No

4.1 Interpretation

- The prediction is **stable** under ± 1 variations in either Betti number (all remain within current 1σ bounds).
- The prediction is **not unique**: multiple topologies yield compatible values with present data.
- Variations of ± 2 or more typically push δ_{CP} outside the allowed range.
- **DUNE precision** (± 5) will be sufficient to distinguish between these topologies.

This sensitivity suggests that a precise DUNE measurement could effectively “measure” the Betti numbers of the internal manifold—a remarkable prospect if confirmed.

5 Discussion: The Mass Hierarchy Clue

Beyond the CP phase, we observe a numerical correlation in charged lepton masses that may be relevant. With $m_e = 0.5110$ MeV and $\phi = (1 + \sqrt{5})/2$:

$$m_\ell(g) = m_e \cdot \phi^{n_g}, \quad n_g \in \{0, 11, 17\} \quad (6)$$

Table 2: Charged lepton masses: prediction vs. experiment

Generation	n_g	m^{pred} (MeV)	m^{exp} (MeV)	Error
e	0	0.5110	0.5110	0.00%
μ	11	101.69	105.66	−3.75%
τ	17	1824.8	1776.9	+2.70%

5.1 Sector Asymmetry Interpretation

We note that:

$$n_\tau - n_\mu = 17 - 11 = 6 = b_2 - b_3 \quad (7)$$

We tentatively interpret $n = b_2 - b_3 = 6$ as the **Gauge-to-Matter Sector Asymmetry**: the imbalance between vector degrees of freedom (b_2) and chiral degrees of freedom (b_3).

Critical Methodological Warning

The exponents $\{0, 11, 17\}$ are **not derived** from the Dirac operator spectrum on G_2 manifolds. The appearance of the golden ratio ϕ in mass scaling has no published theoretical basis.

This pattern is presented as an **intriguing coincidence** that would merit investigation **only if** $\delta_{CP} \approx 198$ is confirmed by DUNE. Until then, it carries no independent evidential weight.

6 Falsification Protocol

The scientific utility of this ansatz lies in its vulnerability to empirical refutation.

Table 3: Falsification criteria

Observable	Prediction	Exclusion Criterion	Test
δ_{CP}	197.8 ± 1	$ \delta_{CP}^{\text{DUNE}} - 197.8 > 10$ at 3σ	DUNE (2030)
Lepton masses	Eq. (6)	Not independently falsifiable	Conditional

6.1 Decision Tree

- ✓ **Confirmation:** DUNE measures $\delta_{CP} = 198 \pm 5 \Rightarrow$ Protocol survives; mass scaling becomes target for theoretical derivation.
- ✗ **Exclusion:** DUNE measures $|\delta_{CP} - 198| > 15 \Rightarrow$ Protocol archived as numerical coincidence. No adjustments, no excuses.

7 Conclusion

We have presented a minimal topological ansatz that makes one falsifiable prediction: $\delta_{CP} \approx 197.8$. The framework explicitly acknowledges:

1. The formula is a phenomenological ansatz, not a rigorous derivation.
2. Multiple topologies yield values compatible with current data.
3. The lepton mass scaling is an unexplained coincidence pending DUNE’s verdict.

This minimalism is intentional. The protocol’s scientific value resides entirely in its capacity for clean empirical refutation. Both confirmation and exclusion advance knowledge: the former would motivate geometric investigation of the mass hierarchy; the latter would eliminate a speculative ansatz without ambiguity.

*“A falsifiable ansatz with topological grounding is a contribution to science;
a derivation that cannot be refuted is a contribution to mythology.”*

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